



CAT 6 FLASHER 12 V VIBRATION TEST REPORT

DATE:25.05.2026

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Part Name	CAT 6 FLASHER 12 V	Application	COMMERCIAL APPLICATION
Customer Part No.	B00 861 544 00 32	Customer	FORCE MOTOR
UCAL Part No.	400230122002	Report No	R&D/CAT6/12VF/DUR/26/032

INTRODUCTION:

UCAL is currently supplying 12V CAT6 Flashers to Force Motor for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 12 V flashers assembly after **Vibration test**.

TEST SPECIFICATION:

The component should be mounted on the actual mounting panel or bracket during the test. Frequency 50 to 500 Hz with a sweep rate of 1 octaves/minute at an acceleration 5 G (P-P). Duration of 10 hours in each three axis. During the test maintain the temperature as per Fig 15. The humidity level of the chamber has to be 90%RH at +80°C.

ACCEPTANCE CRITERIA:

All function of the flasher performs as designed before and after the test.

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:05	O	O	O
S:06	O	O	O

O: OK

X: NG

Δ: To be confirmed

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

 Mr. Rahul Bongarde	 Mr. Karthikeyan N
Prepared By	Approved By

PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	10 V~16 V
2	Test voltage	13.5 V
3	Flash rate(Turn signal/Hazard warning)	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST SETUP:



TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL
		



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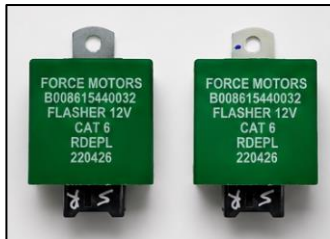
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TEST OBSERVATION:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:05,06	
Turn Signal RH	3 x 21 W Lamp + 1x 2W LED	85	OK
Turn Signal LH	3 x 21 W Lamp + 1x 2W LED	85	OK
Hazard Mode	6 x 21 W Lamp + 2x 2 W LED+	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 2 W LED	170	OK
Duty cycles for RH&LH	3 x 21 W Lamp + 1x 2 W LED	50%	OK
Duty cycles for Hazard Mode	6 x 21 W Lamp + 1x 2 W LED	50%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:05,06	AFTER TEST SAMPLE – S:05,06
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.